



Impact Pulse

Volunteer & Social Service
Activity Management Portal

Connect • Collaborate • Create Impact



Event
Management



Volunteer
Participation



Leadership
Opportunities



Real-time
Notifications



Rewards &
Recognition



Reports &
Insights



Empowering **Volunteers**. Building Stronger **Communities**.



Project Documentation & Development Plan [SRS]



Team Size: 4 Members
By: FusiO(n) Squad

1. Bharani T
2. Yaswanth Kalis P
3. Sri Satish B
4. Yuvaraj N



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1. Problem Statement

Design and develop a comprehensive portal for volunteer and social service activity management that digitizes manual processes, provides role-based access for administrators, staff/organizers, and users/volunteers, automates workflows, generates analytical reports, sends real-time notifications, and integrates securely with institutional systems to improve operational efficiency, transparency, collaboration, and user experience.

Target Users

- Administrators — oversee the entire platform, approve organizers and events, manage system-wide data
- Organizers / Staff — create and manage volunteering events, select team leaders, track attendance
- Volunteers / Users — discover activities, register, track participation history, earn recognition

Major Problems Identified

- Manual, paper-based tracking of volunteer activities and attendance
- No centralized system for approving events or organizers
- Lack of real-time visibility into available activities and seat capacity
- No structured way to recognize or reward volunteer contribution
- Absence of analytics/reporting for institutional accountability

Functional Requirements

- User registration and role-based authentication.
- Event creation, approval, and management.
- Event registration and attendance tracking.
- Leadership application and approval.

- Notification delivery for user activities.
 - Feedback submission and reward management.
 - Administrative management of users, events, and reports.
-

Non-Functional Requirements

- **Performance:** Fast response and concurrent user support.
- **Security:** Secure authentication and role-based access.
- **Reliability:** Accurate and consistent data handling.
- **Usability:** Responsive and user-friendly interface.
- **Scalability:** Supports future growth in users and events.
- **Maintainability:** Modular and well-structured architecture.

Proposed Solution

Overview

The proposed solution is a web-based **Volunteer and Social Service Activity Management Portal** that provides a centralized platform for volunteers, organizers, and administrators to efficiently manage social service events. The system streamlines the entire volunteering process—from event creation and approval to participant registration, attendance tracking, leadership selection, feedback collection, and certificate generation.

The portal eliminates manual paperwork, reduces communication delays, and improves transparency while encouraging community participation through a reward-based system.

Objectives

The proposed system aims to:

- Simplify volunteer registration and participation.
- Enable organizers to efficiently create and manage events.
- Provide administrators with centralized approval and monitoring capabilities.
- Maintain accurate attendance and participation records.
- Recognize volunteers through points, badges, and certificates.
- Improve communication through real-time notifications.
- Generate reports and analytics for better decision-making.

Core Features

Activity lifecycle: draft → published → ongoing → completed → archived, with staff approval gates

Shift/slot-based signups — instead of just "apply," volunteers pick specific time slots (like Calendly for volunteering)

Skill tagging — volunteers tag themselves (first aid, teaching, driving, cooking...) and activities require certain skills

Waitlists — auto-promote from waitlist when someone drops out

Recurring activities — weekly food drives, etc., without recreating each time
Digital consent/waiver forms — especially relevant for institutional/legal compliance angle in the problem statement

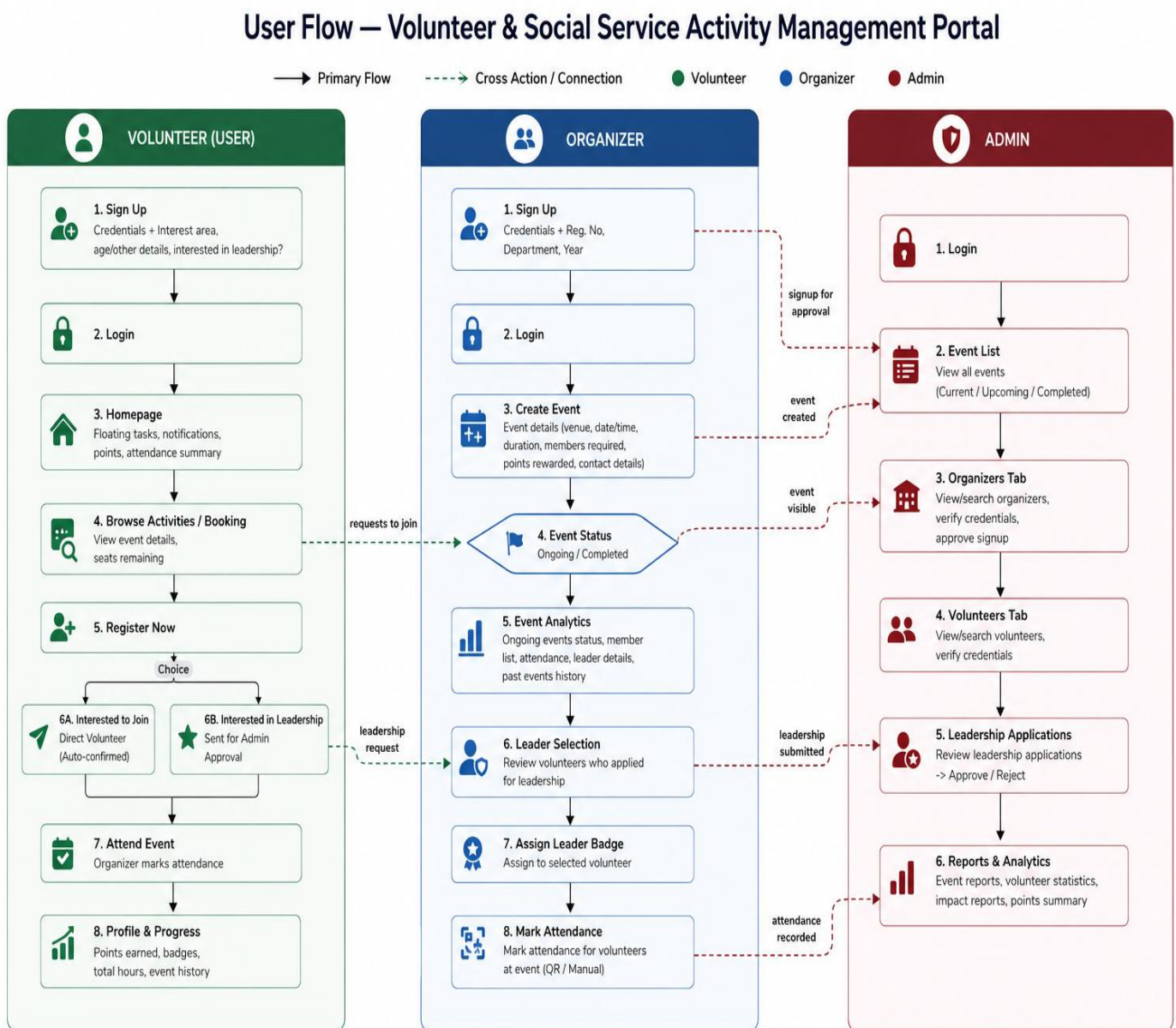
Tech Stack

- Frontend: React (Next.js App Router), Tailwind CSS, shadcn/ui
- Backend: Next.js API Routes / Node.js

- Database & Auth: Supabase (PostgreSQL, Row-Level Security, Realtime, Storage)
- Notifications: Supabase Realtime + email (Resend/Nodemailer)
- Deployment: Vercel

2. System Design & Architecture

2.1 User Flow / Role-Based System Overview



The platform has three primary roles, each with a distinct sign-up flow and dashboard. Signup branches into two paths — Volunteer/User or Organizer — each collecting different credential and interest data before routing to role-specific approval or onboarding.

Role	Access Level	Primary Capabilities
Admin	System-wide	Approve/decline events and organizers; view all organizer & volunteer lists with search; audit and oversight
Organizer / Staff	Event-scoped	Create events; manage ongoing events & history; select team leaders; view member & leader details
Volunteer / User	Self-scoped	Browse & register for activities; track attendance and history; view profile, badges, and points; leaders get quick controls & notifications

2.2 WireFrame

Two signup entry points branch from a single 'Sign Up' action:

A. User / Volunteer Path

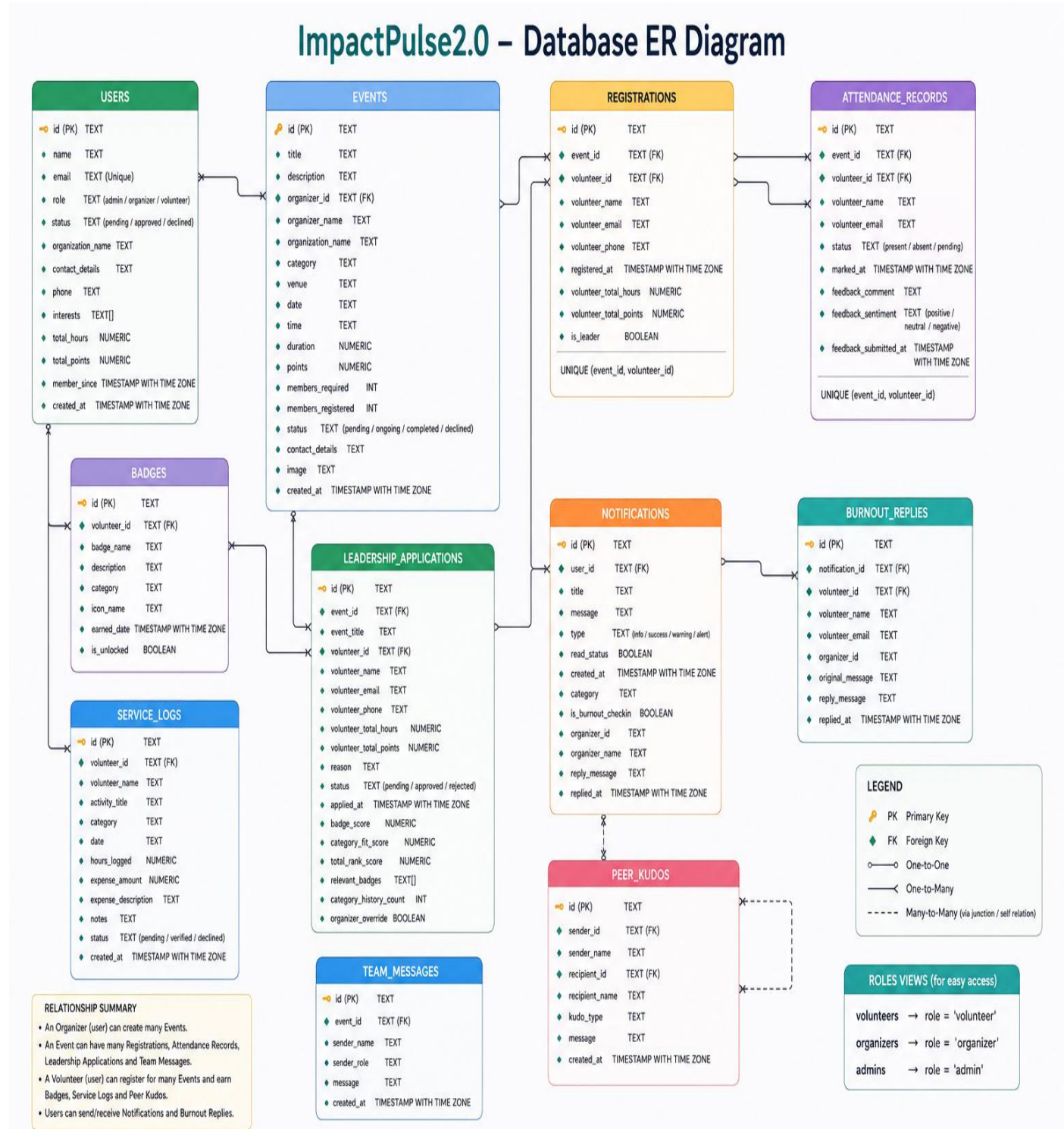
1. User provides credentials (email, password)
2. User fills interest details: interest roles, other requirements (name, age, etc.)
3. User indicates interest in: Leadership OR direct volunteering
4. If interested in leadership → application flagged for admin approval
5. If interested to join directly → routed straight into the Volunteer role

B. Organizer Path

6. Organizer provides credentials
7. Organizer submits organizational details: Registration No., Department, Year
8. Application routed to Admin for approval
9. On approval, organizer gains access to the Organizer dashboard

2.3 Database Design (Entities) + ER DIAGRAM

Core entities required to support the flows above:



Entity	Key Fields
Users	id, name, email, password_hash, role (admin/organizer/volunteer), age, reg_no, department, year, status (pending/approved)
Events	id, organizer_id, title, description, members_required, timing, date, duration, venue, contact_details, points_rewarded, status (approved/ongoing/completed/cancelled), available_seats
Registrations	id, event_id, user_id, registration_date, attendance_status, role_in_event (member/leader)
LeadershipApplications	id, user_id, event_id, status (pending/approved/declined), badge_awarded
Notifications	id, user_id, message, type, is_read, created_at
Feedback	id, event_id, user_id, comments, rating
Badges/Points	id, user_id, total_points, member_since, badges_earned

2.4 API Planning (High-Level)

API Endpoint Structure

Authentication

- POST** /api/auth/signup
 Register a new user with role-based access (*Volunteer / Organizer*).
- POST** /api/auth/login
 Authenticate user and generate access token.

Event Management

- GET** /api/events
 Retrieve all events with optional filters (*status, organizer, search*).

- **POST** /api/events
Create a new event (*Organizer only*).
 - **PATCH** /api/events/:id/status
Approve, decline, or cancel an event (*Admin only*).
-

Event Registration

- **POST** /api/events/:id/register
Register a volunteer for an event.
 - **PATCH** /api/events/:id/attendance
Mark volunteer attendance (*Organizer / Assigned Leader*).
-

Leadership Management

- **POST** /api/leadership/apply
Submit a leadership application for an event.
 - **PATCH** /api/leadership/:id/approve
Approve or reject a leadership application (*Admin only*).
-

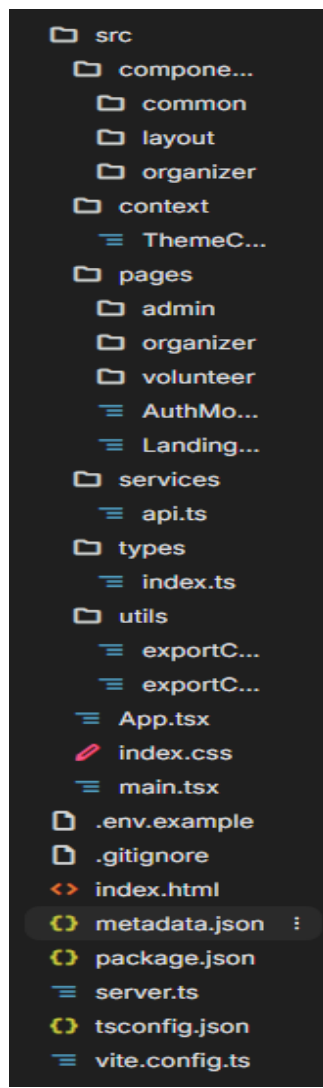
Notifications

- **GET** /api/notifications
Retrieve notifications for the authenticated user.

Administration

- **GET** /api/admin/organizers
Retrieve and search organizer records with registration details.
- **GET** /api/admin/volunteers
Retrieve and search volunteer records with profile details.

2.5 Project Folder Structure



3. Feature Breakdown by Role

3.1 Organizer Dashboard

Tab 1 — Create Event

- Event title & description
- Members required
- Timing & date, duration
- Venue and contact details
- Points rewarded on completion
- Available seats remaining (auto-tracked)

Tab 2 — My Events

- Ongoing Events → shows member & leader details
- History → past events with status and full event details

Tab 3 — Leader Selection

- Shows all volunteers who applied for leadership on the organizer's events
- Organizer reviews and assigns the 'Leader' badge to selected volunteers

3.2 Admin Dashboard

Tab 1 — Event List

- Current Events → Approve / Decline pending events
- Status tracking: Approved → Ongoing → Completed / Cancelled

Tab 2 — Organizer's Tab

- Full searchable list of organizers with their credentials

Tab 3 — Volunteer's Tab

- Full searchable list of volunteers with their credentials

3.3 Volunteer / User Dashboard

Tab 1 — Homepage

- Floating task shortcuts
- Notifications feed
- Points summary widget

Tab 2 — Booking

- Available Activities → browse and register for open events
- History → previously attended events with attendance record

Tab 3 — Profile

- Personal details
- Badges earned and total points
- Member-since date and total volunteering hours

4. Alignment with Judging Criteria (Innovation)

- Role-based workflow (Admin → Organizer → Volunteer) mirrors real institutional processes, directly answering the brief's transparency and accountability requirements
- Leadership application/approval flow adds a growth path for volunteers, differentiating from a plain signup-and-attend model
- Badge, points, and hours tracking give the portal a gamified retention layer beyond the base requirements

5. Summary

- **VolunteerHub** digitizes the full volunteer management lifecycle — from event creation and approval, through discovery and registration, to attendance and recognition — with an AI layer that makes matching and reporting genuinely intelligent rather than cosmetic.
- The three-person team split by feature domain (Admin/Auth, Volunteer-facing, AI/Analytics) allows parallel development across the 48-hour window, with a clear checkpoint at Hour 24 for a fully demoable core product.

USER FLOW DIAGRAM – VOLUNTEER & SOCIAL SERVICE ACTIVITY MANAGEMENT PORTAL

